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Subsidies Upgraded for the Purchase of Clean Energy Vehicles toward the Realization of GX in the Automobile Sector

(Provisional translation)

Decarbonization

Global warming

(English ver.) 2024-08-22

Sustainability

Electric Vehicle (EV)

Electrified vehicle

Post



Japan has set a goal for all sales of new passenger vehicles to be electrified vehicles by 2035. Electrified vehicles include electric vehicles (EVs), fuel cell vehicles (FCVs), plug-in hybrid electric vehicles (PHEVs), and hybrid electric vehicles (HEVs.) As they emit less CO2 than conventional cars with internal-combustion engines (ICE vehicles), various kinds of electrified vehicles have been brought to the market. HEVs accounted for about 50% of the domestic sales of new passenger vehicles in FY2023, whereas EVs, FCVs and PHEVs combined had only a 3.5% share. Quite a few people may have the notion that these three types of electrified vehicles are expensive and exclude them from their options for purchasing new cars. In order to popularize these three types of electrified vehicles, we would like to encourage them to take advantage of the Subsidies to Promote the Introduction of Clean Energy Vehicles (a.k.a. CEV Subsidies). With these subsidies granted, people can not only enjoy low purchase prices but also contribute to efforts toward achieving net-zero GHG emissions. This article highlights CEV subsidies that have been upgraded from FY2024.

To achieve the goal of all sales of new passenger vehicles being electrified vehicles by 2035

CO2 emissions from the automobile sector account for about 16% of Japan's total CO2 emissions. It is essential to promote and popularize clean energy vehicles toward achieving the goal of net-zero GHG emissions by 2050.

As mentioned before, Japan has set a goal for all sales of new passenger vehicles to be electrified vehicles by 2035. Toward achieving this goal, the country has been offering CEV subsidies to support the new purchase of clean energy vehicles.

Creating GX markets in the automobile sector through CEV subsidies

The concept of CEV subsidies has been restructured from FY2024 with major changes made to the evaluation method to determine the amounts of subsidies.

In the past, the amounts of subsidies were determined based solely on vehicle performance, such as the electric mileage (how far the car can go with one kilowatt-

hour) and mileage per charge (how far the car can go with the installed batteries fully charged.) The evaluation method has been changed from 2024 to comprehensively assess car makers' efforts based on the values necessary for the realization of GX in the automobile sector. GX means green transformation from fossil fuel (coal and oil)-centered industrial and social structures to clean energy-centered ones with net zero emissions.

What specifically then are the values necessary for the realization of GX in the automobile sector? In order to achieve GX in the automobile sector, it is essential to create an environment in which people can drive their electrified cars safely and with peace of mind. To this end, charging infrastructure must be developed and after-sales services should also be improved, not to mention the improvement of vehicle performance. Furthermore, life-cycle CO2 emissions from manufacturing to disposal need to be reduced wherever possible in addition to emissions while travelling. It is also expected that electrified vehicles will serve as emergency power sources in disasters. These aspects have been added to the values necessary for the realization of GX in the automobile sector.

This aims for the creation of GX markets in the automobile sector by accelerating efforts by car makers toward creating an environment in which electrified vehicles are utilized sustainably.

Values necessary for the realization of GX in the automobile sector

1. Improving product performance

- Longer electric mileage and cruising distance
- Eligible for the Top Runner System under the Act on Rationalizing Energy Use

2. Creating a user-friendly environment to ensure safety and peace of mind

- Developing charging infrastructure
- Improving after-sales service and fostering human resources for maintenance
- Strengthening cyber security for vehicles

3. Ensuring sustainability over the entire life cycle

- Reducing CO2 emissions over the entire life cycle
- Proper implementation of reuse and recycling, and efficient resource utilization

4. Contributing to other fields through the utilization of automobiles

- Equipped with external power outlets to cooperate with local communities in disaster response

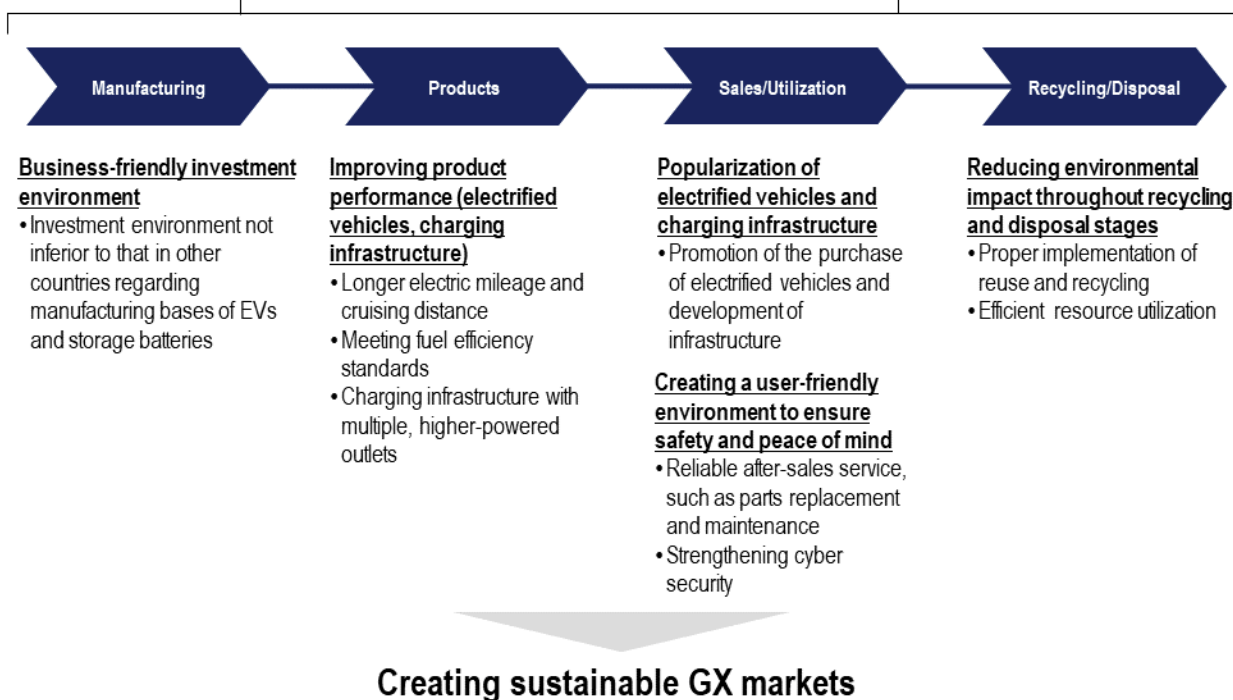
Enlarged View

Ensuring sustainability over the entire life cycle

- Using green materials, improving production processes

Contributing to other fields through the utilization of automobiles

- Energy management, cooperation in disasters



Enlarged View

Expanded budget for subsidies, such as a maximum of 850,000 yen for an EV and 550,000 yen for a PHEV

The total budget for CEV subsidies that can be applied for in FY2024 has been increased to 129.1 billion yen. The subsidy amounts are calculated based on the above values that respective vehicle models are expected to create.

The maximum amount of the subsidy for each vehicle category is as follows:

850,000 yen for an EV

550,000 yen for a light automobile EV

550,000 yen for a PHEV

2,550,000 yen for an FCV

New vehicles that are registered with inspection on and after April 1, 2024, are eligible for the new subsidy scheme.

Vehicle category	Subsidy amount
EV	150,000 up to 850,000 yen
Light automobile EV	150,000 up to 550,000 yen
PHEV	150,000 up to 550,000 yen
FCV	Up to 2,550,000 yen

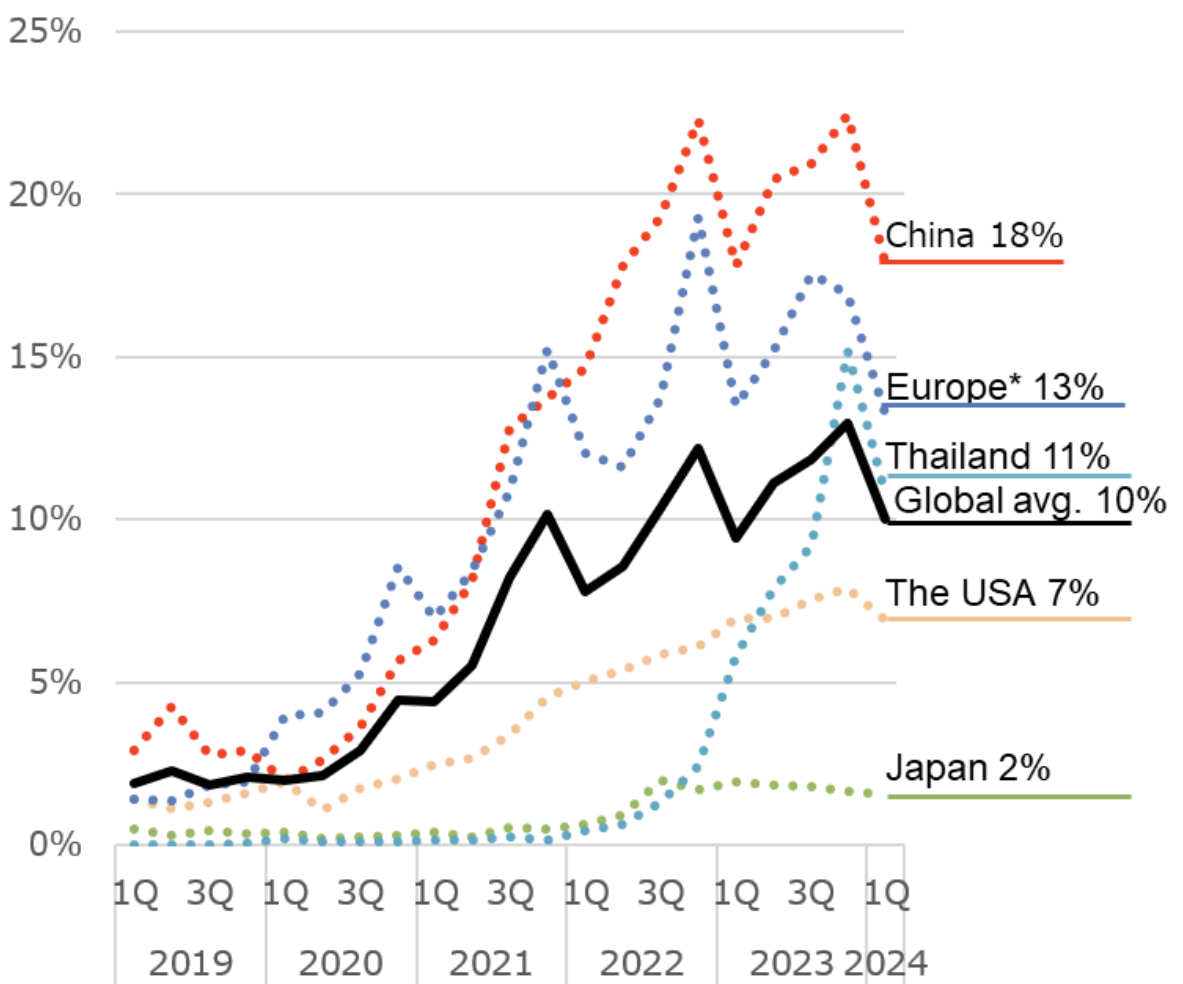
Note: If an EV, a PHEV or an FCV has a manufacturer's recommended retail price of 8,400,000 yen or more, the calculated subsidy amount is reduced by 20% with a coefficient of 0.8. The old system for calculating subsidies applies to ultra-compact mobility vehicles, mini cars, and electrified two-wheel vehicles.

Enlarged View

Japan strategically pursues various possibilities for electrified vehicles, rather than focusing only on EVs

The following graph shows the ratios of EV sales in major countries and regions. The ratio of EV sales in Japan is as low as 2% while those in China and in Europe are 18% and 13% respectively.

Changes in the ratios of EV sales in major countries and regions



(Source) Marklines

*The figure for Europe represents three countries, namely the UK, France, and Germany

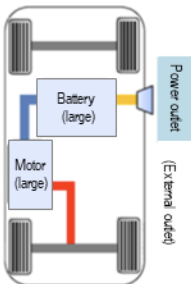
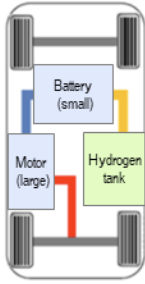
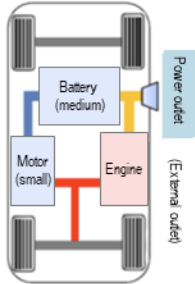
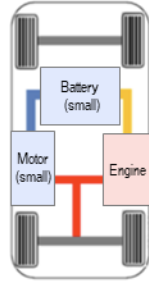
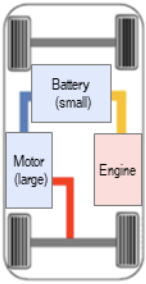
Enlarged View

Japan adopts a policy of promoting not only EVs but also other types of electrified vehicles, such as FCVs, PHEVs and HEVs.

Each type of electrified vehicle has its own strengths and weaknesses in terms of cruising distance, charging time/filling time, cost, and CO2 emissions. Taking into account such strengths and weaknesses, Japan aims to utilize each type of electrified vehicle in the right place in society, such as light automobile EVs for short distance travelling and FCVs for long distance bus routes.

Strengths and weaknesses of each type of electrified vehicle

■ Power
 ■ Electricity supply to batteries
 ■ Electricity supply to motors

	EV/BEV	FCV	PHEV	HEV		
				Toyota (Prius)	Nissan (e-Power)	
Target for electrification	100 % new car sales by 2035					
	USA: 50% of new car sales by 2030/China: 50% of new car sales by 2035					
	Japan: 100% of new car sales by 2035					
Structure						
Strengths	No CO2 emissions while moving	No CO2 emissions while moving	No CO2 emissions in electrified mode	Better fuel mileage than conventional cars with ICEs		
		Long cruising distance	Engine can be utilized when out of electricity			
		Short filling time				
Weaknesses	High cost	Higher cost than Evs	CO2 emitted while moving in engine mode	CO2 emitted while moving if not as much as gasoline-fueled cars		
	Short cruising distance					
	Long charging time	High cost of filling infrastructure	High Cost			
	Large CO2 emissions in manufacturing batteries					

Note: The definition of a ZEV (zero emission vehicle) varies from region to region.
 (Source) Created by the Ministry of Economy, Trade and Industry based on published information

Enlarged View

Since there is no perfect technology at the moment for achieving GX in the automobile sector, R&D for each type of electrified vehicle still needs to be advanced. For instance, in order to decarbonize HEVs, synthetic fuel (e-fuel) is under research toward commercialization by the early part of the 2030s. Furthermore, next-generation batteries are being developed for EVs and PHEVs. A large-scale hydrogen supply chain needs to be built for the popularization of FCVs.

The Green Innovation Fund is being utilized to accelerate the development of these technologies. We are taking various policy measures for the realization of GX in the automobile sector, promoting innovation on the supply side, and offering subsidies on the demand side for the purchase of electrified vehicles toward their solid popularization in society.

Divisions in Charge

About this article

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About the Special Contents

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